Kenya as East Africa's low-carbon industry hub: executive summary

Question. Should Kenya position itself as East Africa's hub for energy-intensive, low-carbon industry (data centres first), and what would it take?

Answer. Yes, but in sequence. Kenya's clean power is a real advantage, but the grid has almost no spare capacity at the 6–10pm peak. Fix evening supply first, pilot with firmed and flexible load, and scale only while the reserve margin stays at 10% or more.

WHAT THE ANALYSIS SHOWS

1. **The clean-power advantage is real.** Renewables supplied about 79% of generation in July–December 2025, and geothermal alone supplied 40%.
2. **The binding constraint is evening capacity, not carbon.** The reserve margin fell from 20.7% in January 2026 to 3.3% in June 2026, against a 20–35% norm. Microsoft and G42's planned $1bn Olkaria data centre is in limbo over power. Even with no hub, Base supply turns negative in 2029 (−2.8% in 2030).
3. **The prize is large in investment, small in jobs.** A targeted hub (12% of Africa's 2030 data-centre capacity) means 222 MW of IT load, a $300m a year infrastructure-fee pool and $1.4bn of build investment by 2030, but only about 950 operating jobs.
4. **The hub is not the cause of the shortage.** It adds 155 MW of evening load, about a fifth of the 754 MW firm-capacity gap that already exists to reach a 15% margin by 2030.
5. **Economics are marginal.** A 44 MW reference campus earns a 9.1% IRR against a 10% hurdle (NPV −$15.8m). The break-even fee is $156 per kW-month against $150 assumed. Each extra $1m per MW of build cost removes $44m of NPV. Kenyan grid power costs a tenant about $26 per kW-month more than an Ethiopia-linked price, a 10% handicap on the all-in bill.
6. **Colocation is not the best anchor.** In a weighted screen of seven industries, agro-processing and night-shift heavy industry score highest (4.30 and 4.10). Cloud and colocation data centres rank sixth (3.15) because they cannot flex load.
7. **The gap can be closed.** Five levers add 805 MW of firm evening capacity (1.07 times the gap) for about $287m a year and lift the 2030 reserve margin to 16.5%. Data centres supply 19% of that themselves through on-site storage.

RECOMMENDATION

- **Phase 1, now to month 12 (Gate A: auction and storage at financial close).** Run the first auction round, secure the 100 MW KenGen battery, agree a second Ethiopia import tranche, require new data centres to bring their own evening storage, and relax night-tariff rules.
- **Phase 2, months 12 to 36 (Gate B: reserve margin at or above 10% after the pilot).** Connect two campuses, about 100 MW IT in total, and open a zone for agro-processing and night-shift industry on the time-of-use tariff.
- **Phase 3, month 36 onward (Gate C: about 800 MW of firm evening capacity delivered).** Scale to the targeted hub of about 220 MW IT, tranche by tranche.

CONFIDENCE AND LIMITS

Several grid inputs come from press reports quoting Kenya Power and EPRA briefings, not primary datasets. Colocation pricing, utilisation, operating cost and the discount rate are analyst assumptions, because no public Kenyan price data was found. The industry screen scores are judgement. Treat the results as a structured starting point, and confirm inputs against EPRA statistics before relying on them.

Source for every figure: `model/kenya_green_hub_model.xlsx`. Method, hypotheses and source register: `docs/methods_appendix.md`.